

Exercises in Marine Ecological Genetics

08. Selection

- Window-based genome scans
- F_{ST} outlier loci
- Candidate genes under selection
- Haplotype-based selection tests

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<https://github.com/mhelmkampf/meg26>



Today's objectives

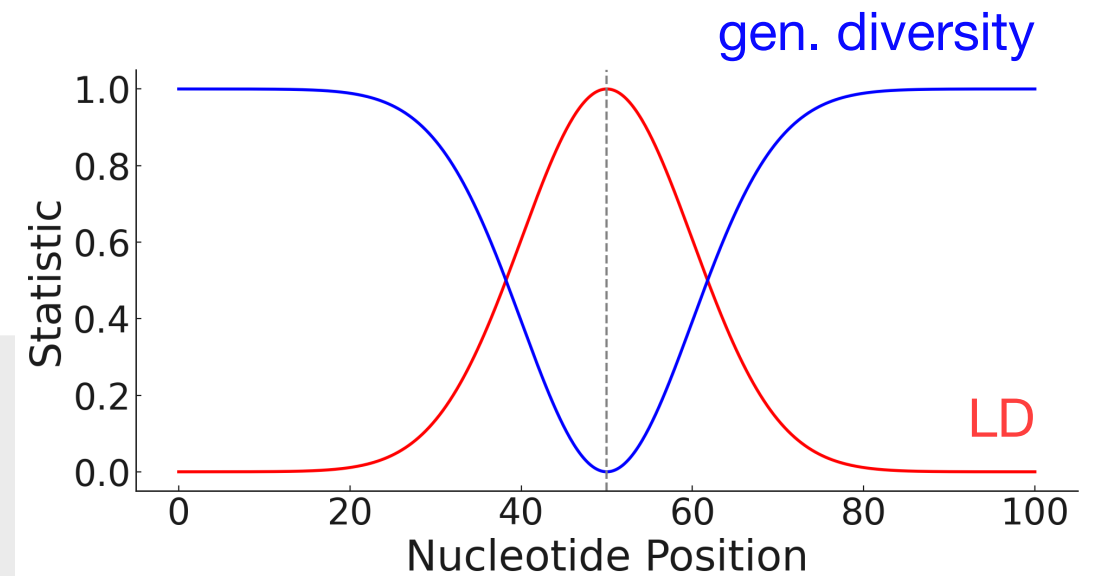
By the end of this session, you will be able to:

- Perform a **genome-wide scan for selection**, choosing suitable windowing parameters and calculating population differentiation statistics (e.g., F_{ST})
- Detect and identify **F_{ST} outlier loci** as potential targets of selection
- Link outlier loci to **candidate genes** by querying gene annotation files (GFF) and functional databases
- Evaluate the strengths and limitations of F_{ST} -based scans relative to **haplotype-based methods** for detecting selection

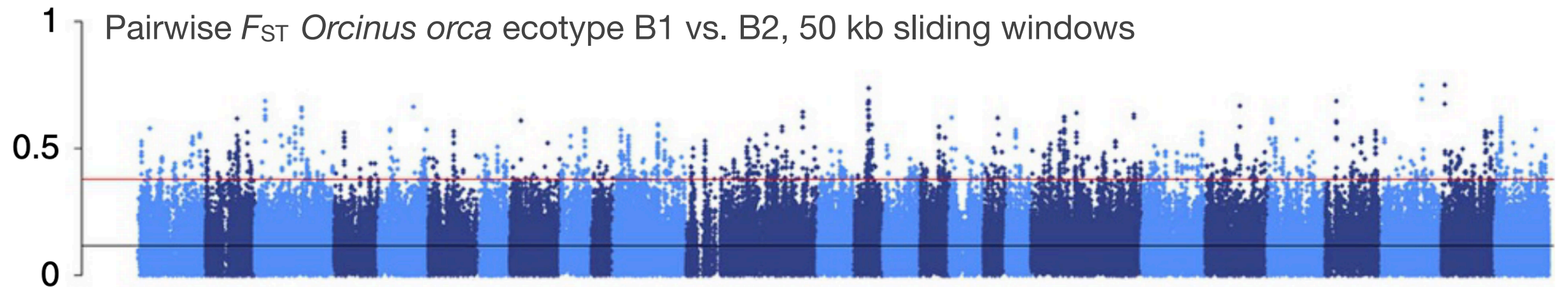
Detecting selection

Positive selection influences genetic variation by:

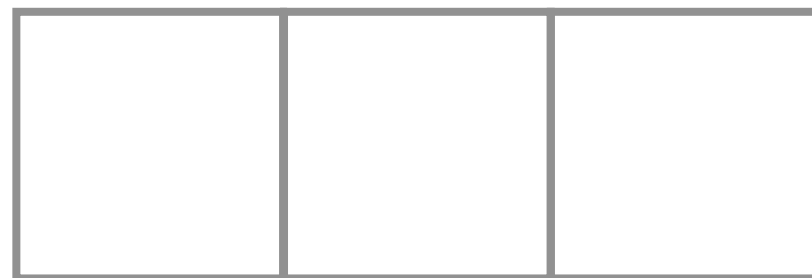
- Reducing genetic diversity (e.g. π)
- Increasing linkage disequilibrium
- Skewing allele frequencies / abundance of rare alleles (Tajima's D)
- **Increasing population differentiation (F_{ST})**
- Extending homozygous haplotypes (iHS, EHH)



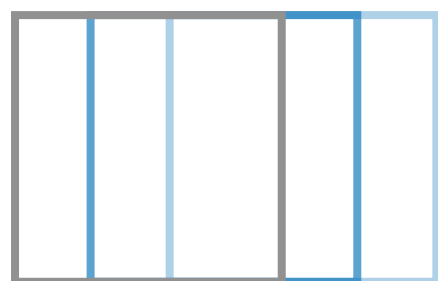
Genome-wide scans using windows



Footo et al. 2016, *Nature Communications*



... Non-overlapping windows



... Overlapping / sliding windows



Window size

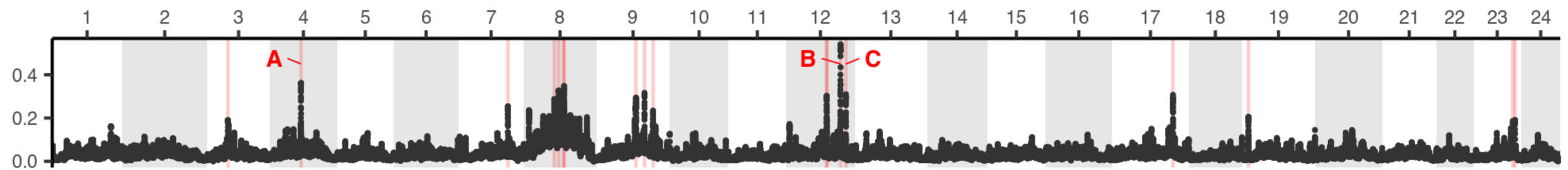


Increment / step size

F_{ST} scan across whole genome

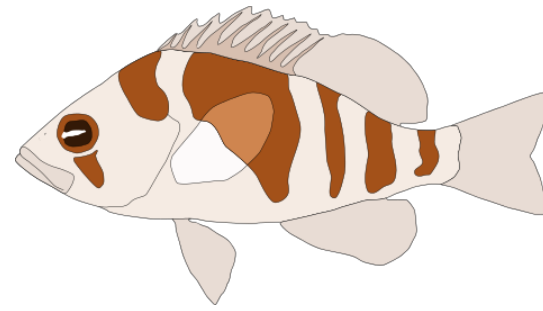
VCFTools, 50 kb sliding windows, joint F_{ST} of 14 Caribbean hamlet populations

Red = 99.8th percentile

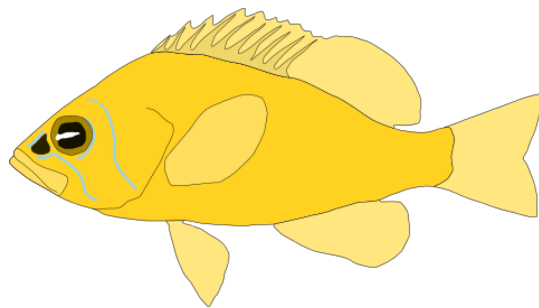


Hench et al. 2022, PNAS

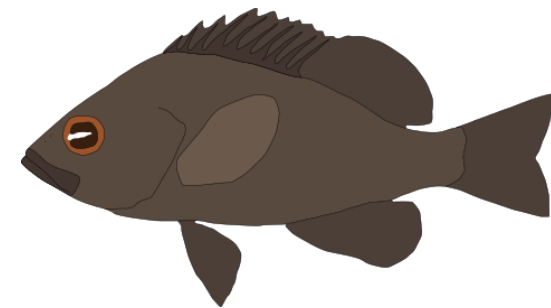
Example data



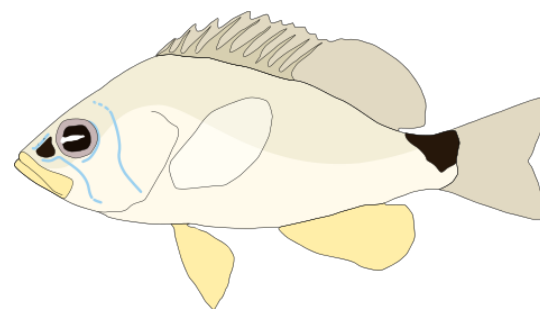
H. puella



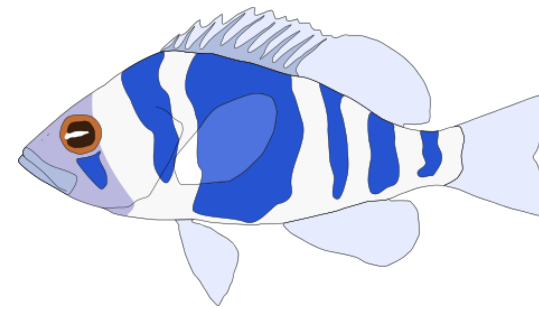
H. gummigutta



H. nigricans

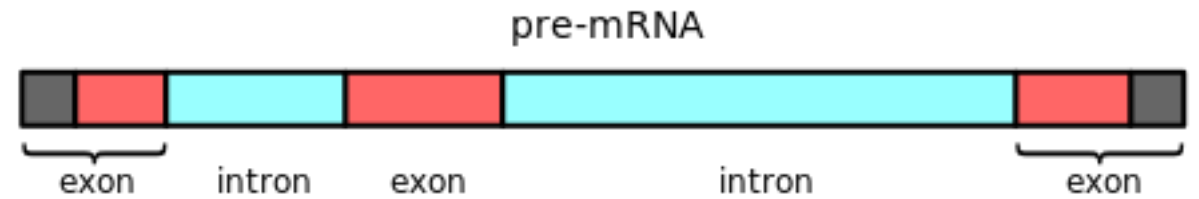


H. unicolor



H. indigo

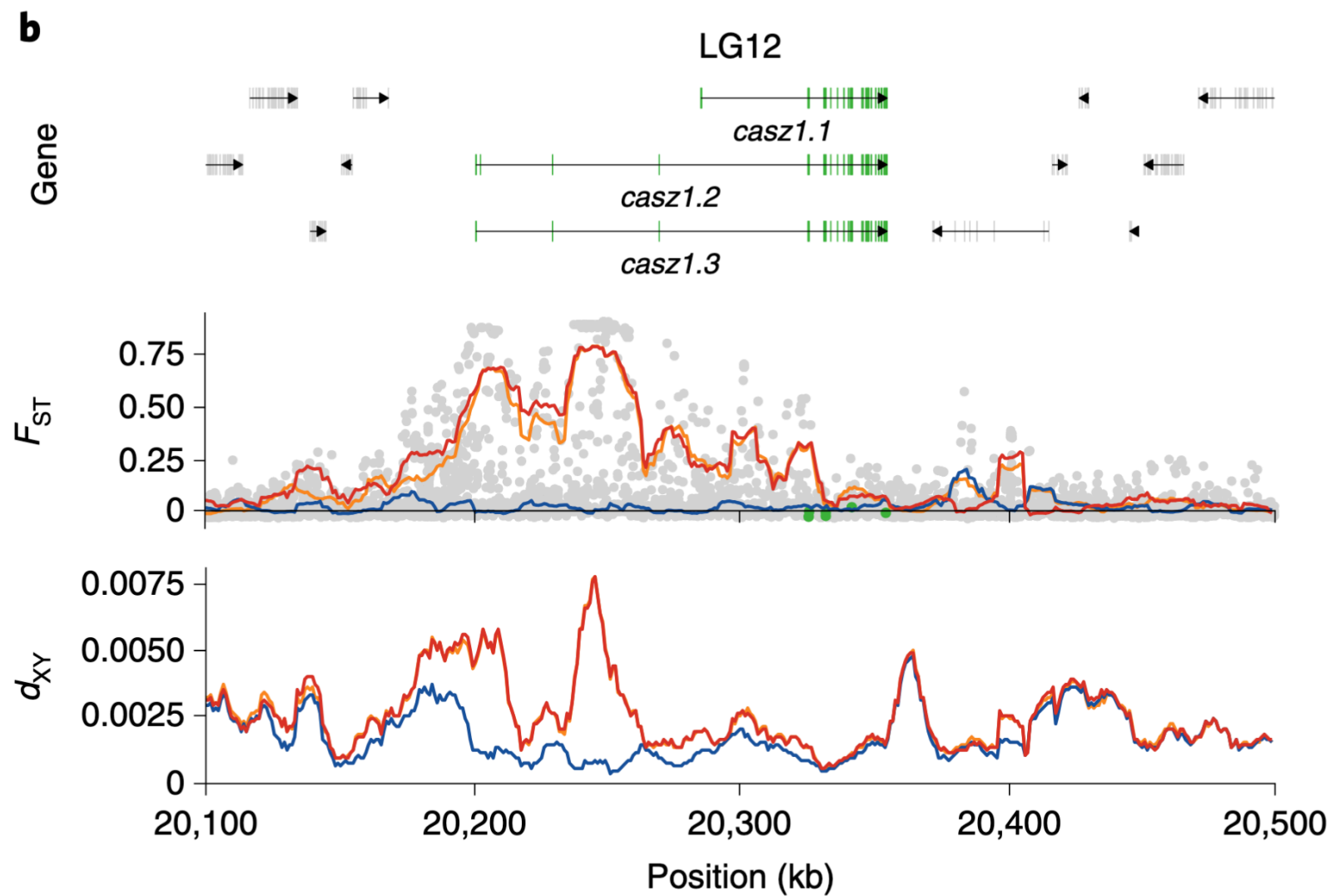
General Feature Format (GFF)



```
##gff-version 3;Parentgene=;Parentgenename=
Contig49      .          gene          406          8055          .          +
.            ID=HPv1G000000000001;...;Parentgenename=ZNF484
Contig49      .          mRNA          406          8055          .          +
.            ID=HPv1M000000000001;...;Parentgenename=ZNF484
Contig49      .          exon          406          700          .          +
.            ID=HPv1E000000000001;...;Parentgenename=ZNF484
Contig49      .          exon          5740         5843          .          +
.            ID=HPv1E000000000002;...;Parentgenename=ZNF484
Contig49      .          exon          5993         6110          .          +
.            ID=HPv1E000000000003;...;Parentgenename=ZNF484
Contig49      .          exon          7271         8055          .          +
.            ID=HPv1E000000000004;...;Parentgenename=ZNF484
...
```

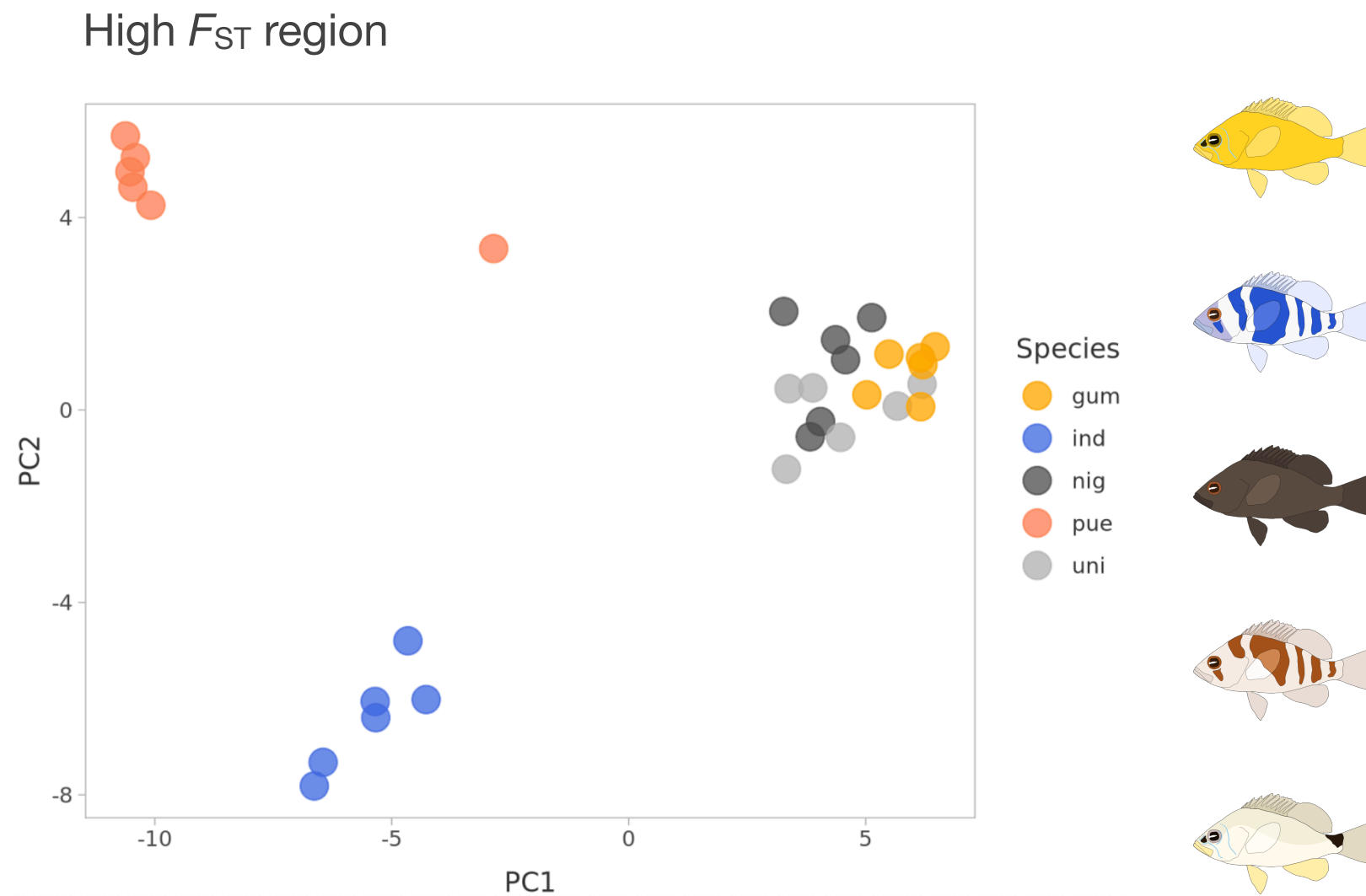
Image from biologyonline.com/

Genome-wide scans centered on *casz1* outlier region

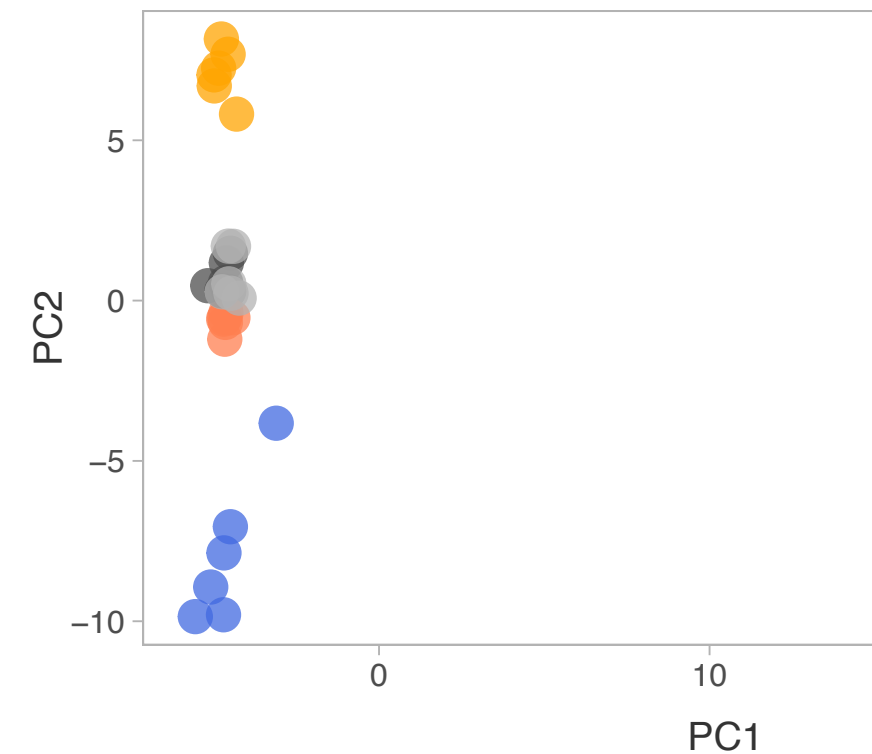


Hench et al. 2019, *Nat Ecol Evol*

PCA of F_{ST} outlier region on LG12

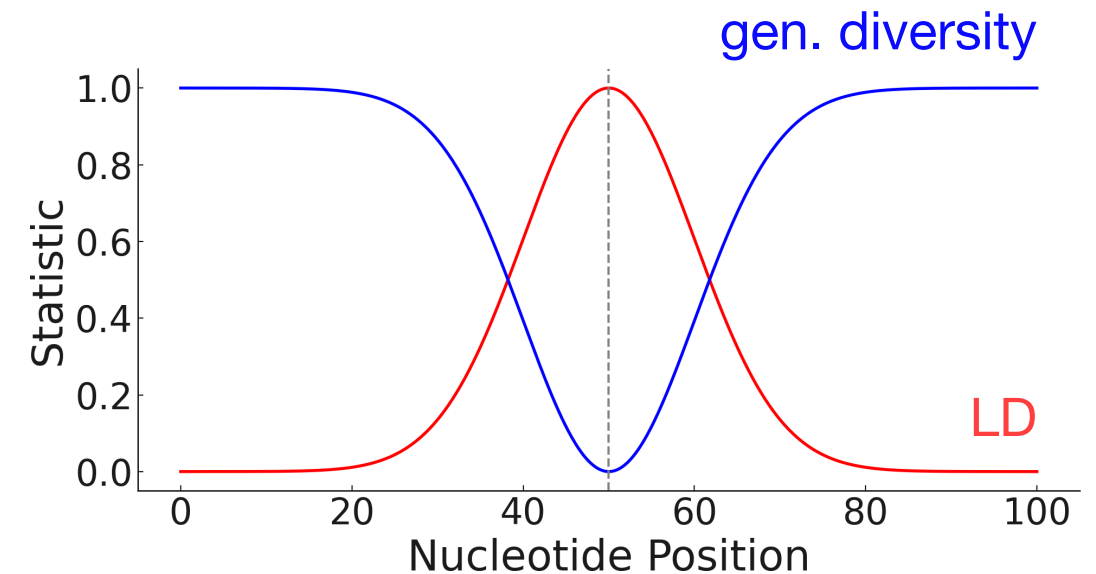


Whole chromosome LG12

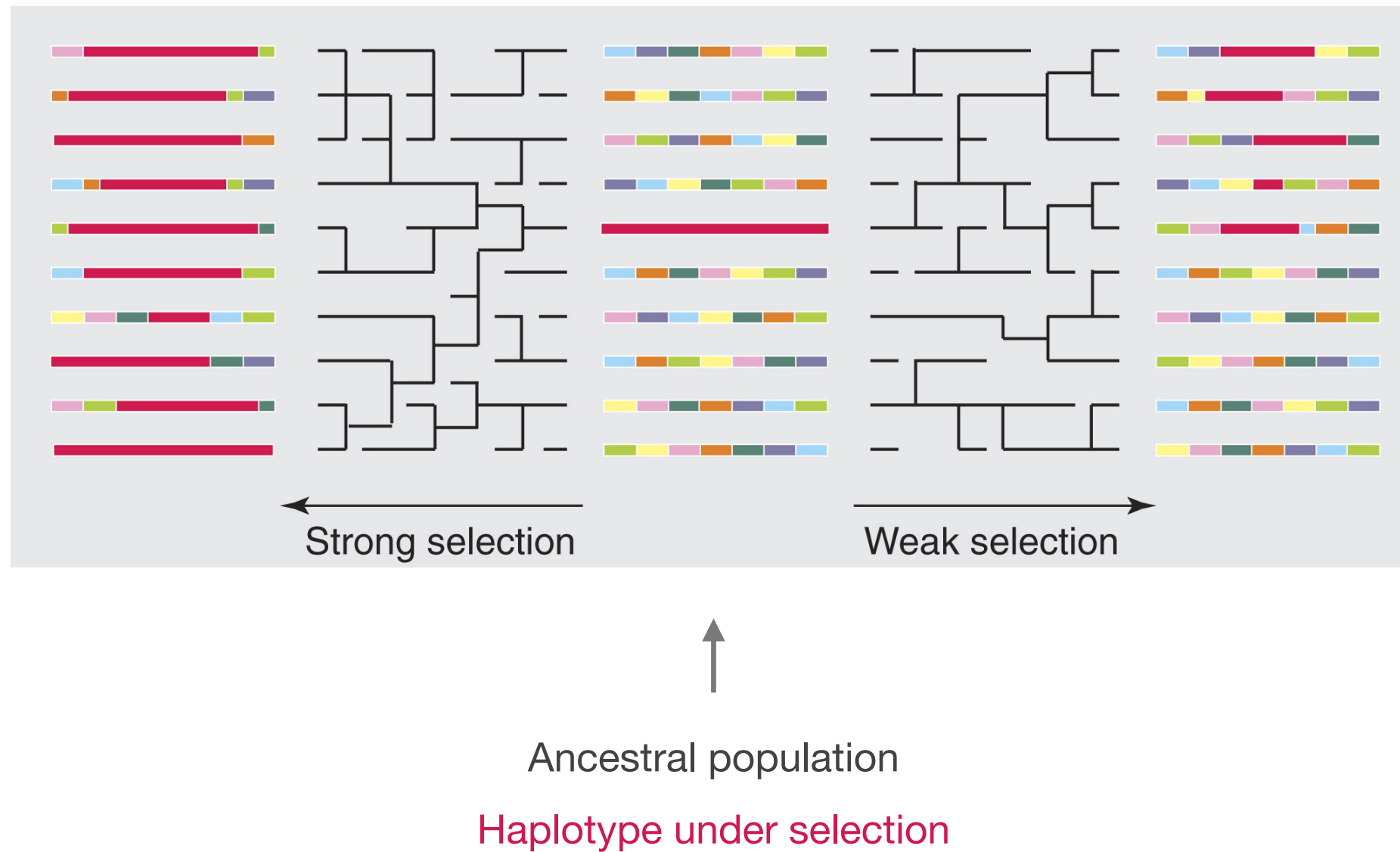


Haplotype-based methods

- Positive selection increases the frequency of a beneficial mutation in the population
- Nearby loci “hitchhike” due to linkage disequilibrium, faster than recombination can break up the association (**selective sweep**)
- As the haplotype rises in frequency, more individuals carry the same extended haplotype around the selected site (**extended haplotype homozygosity**)
- Haplotype-based methods (e.g. iHS, XP-EHH) detect this pattern of long, high-frequency homozygous haplotypes as a signature of **recent positive selection**
- Longer homozygous haplotypes = stronger evidence for selection



Haplotype-based methods



Gibson 2007, *Current Biology*

Phylogeny of *casz1* region

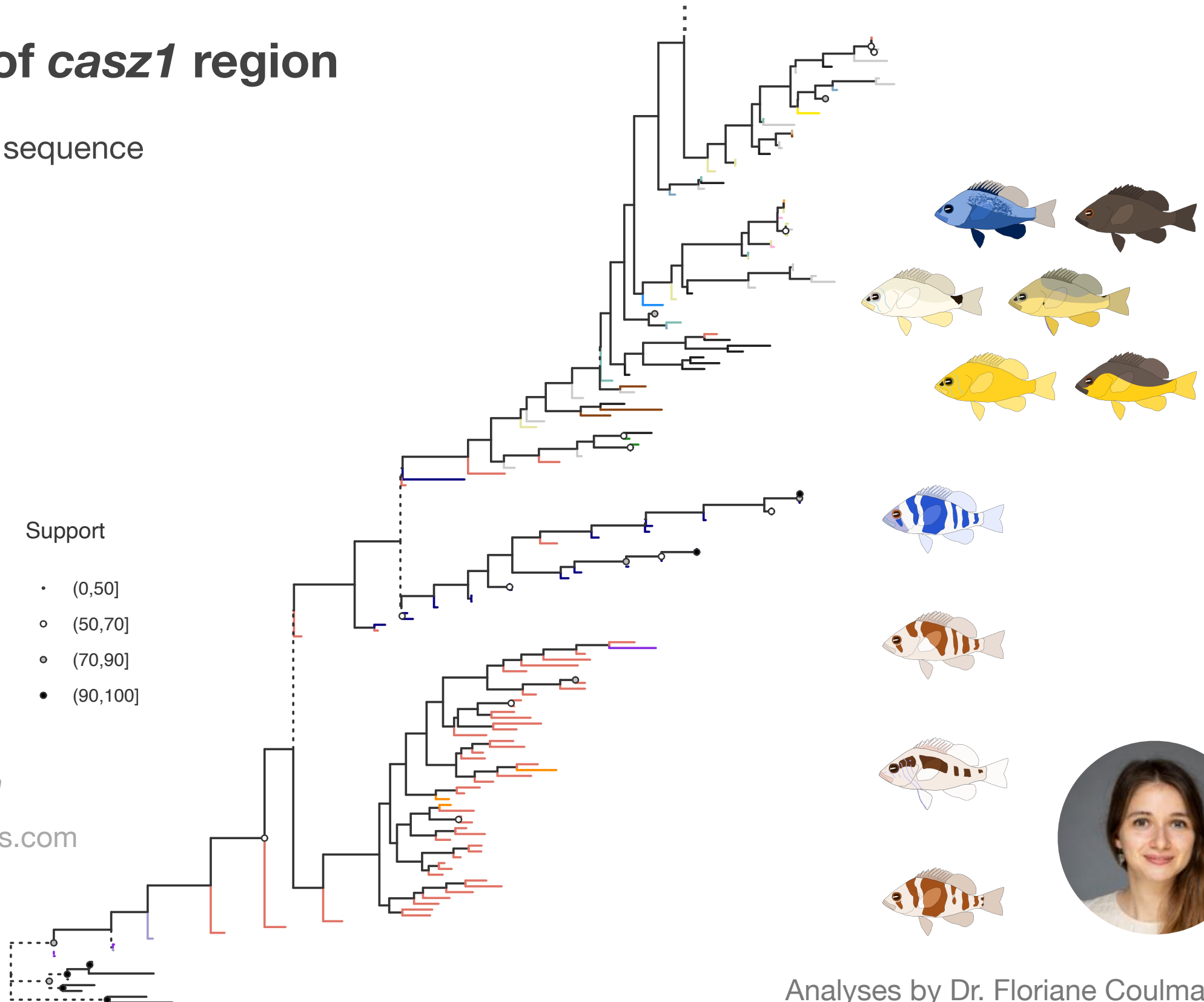
RAxML | ~ 200 kb sequence

Support

- (0,50]
- (50,70]
- (70,90]
- (90,100]

Serranus tortugarum

Image by whitecorals.com



Analyses by Dr. Floriane Coulmance

Selection

Summary

[illegible]

```
vcftools \                                     ### Fst-based scan along 50 kb windows
--gzvcf input.vcf.gz \
--weir-fst-pop pop_1.txt \
--weir-fst-pop pop_2.txt \
--fst-window-step 5000 \
--fst-window-size 50000 \
--stdout 1> Fst_50k.tsv 2> Fst_50k.log
```

[illegible]

```
scan_hh()      ### Scan for EHH (Extended Haplotype Homology), rehh package
```

```
manhattanplot() ### Create Manhattan plot, rehh package
```

- Sliding-window scans help identify **regional genomic signals of selection** while reducing noise from individual markers
- Elevated F_{ST} values highlight candidate loci that **may be influenced by divergent selection**, but alternative explanations need to be evaluated
- **Functional gene annotation** transforms statistical signals into biologically meaningful hypotheses about adaptation
- F_{ST} - and haplotype-based methods provide complementary insights because they detect different signatures and timescales of selection